

```
Windows PowerShell
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PS C:\Users\guyma> cd C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> git switch jeanne
Already on 'jeanne'
Your branch is up to date with 'origin/jeanne'.
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> |
```

1. La fonction qui lit le JSON, c'est **load_instruments()**.
2. Celle qui lit le CSV, c'est **load_prices()**.
3. Celle qui sélectionne un ticker, c'est **filter_prices()**.
4. Celle qui coordonne l'exécution, c'est **main()**.
5. Les fichiers d'entrée sont stockés dans **data/sample/**

```

1  from pathlib import Path
2  import csv
3  import json
4
5  DATA_DIR = Path("data/sample")
6  LOOKBACK_LABEL = "1 month"
7  INTERVAL_LABEL = "Daily"
8
9  def load_instruments():
10     with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
11         return json.load(file)
12
13
14  def load_prices():
15     with open(DATA_DIR / "prices.csv", encoding="utf-8") as file:
16         return list(csv.DictReader(file))
17
18
19  def filter_prices(prices, ticker):
20     return [row for row in prices if row["ticker"] == ticker]
21
22
23  def get_first_close(prices):
24     return float(prices[0]["close"])
25
26
27  def get_last_close(prices):
28     return float(prices[-1]["close"])
29
30
31  def display_configuration():
32     print("Market configuration")
33     print(f"Period : {LOOKBACK_LABEL}")
34     print(f"Interval : {INTERVAL_LABEL}")
35
36
37  def main():
38
39     instruments = load_instruments()
40     prices = load_prices()
41     instrument = instruments["instrument"]
42     benchmark = instruments["benchmark"]
43     instrument_prices = filter_prices(prices, instrument["ticker"])
44     benchmark_prices = filter_prices(prices, benchmark["ticker"])
45     print("=== MarketPulse ===")
46     print()
47     display_configuration()
48     print()
49     print("Instrument")
50     display_market_summary(instrument, instrument_prices)
51     print()
52     print("Benchmark")
53     display_market_summary(benchmark, benchmark_prices, show_currency=False)
54
55
56  if __name__ == "__main__":
57     main()

```

```
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PS C:\Users\guyma> cd C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> git switch jeanne
Already on 'jeanne'
Your branch is up to date with 'origin/jeanne'.
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> python src/main.py
=== MarketPulse ===

Instrument
AAPL - Apple Inc.
Last price: 266.20 USD

Benchmark
SP500 - S&P 500
Last level: 6742.00

Period: 1 month
Interval: Daily

Observations
AAPL: 21
SP500: 21
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryharders> python
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr  8 2025, 12:21:36) [MSC v.1943 64
bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> import json, csv
>>> instruments = json.load(open("data/sample/instruments.json"))
>>> type(instruments)
<class 'dict'>
>>> instruments.keys()
dict_keys(['instrument', 'benchmark'])
>>> instruments["instrument"]["ticker"]
'AAPL'
>>> instruments["benchmark"]["name"]
'S&P 500'
>>> prices = list(csv.DictReader(open("data/sample/prices.csv")))
>>> type(prices)
<class 'list'>
```

```
<class 'list'>
>>> type(prices[0])
<class 'dict'>
>>> prices[0]["close"]
'250.00'
>>> type(prices[0]["close"])
<class 'str'>
>>> float(prices[0]["close"])
250.0
>>> exit()
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> notepad src/main
.py
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> code src/main.py
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> python src/main.py
=== MarketPulse ===

Market configuration
Period : 1 month
Interval : Daily

Instrument
AAPL - Apple Inc.
Observations : 21
First close : 250.00 USD
Last close : 266.20 USD

Benchmark
SP500 - S&P 500
Observations : 21
First close : 6600.00
Last close : 6742.00
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> cat data/sample/instruments.json
{
  "instrument": {
    "ticker": "AAPL",
    "name": "Apple Inc.",
    "currency": "USD",
    "market": "NASDAQ"
  },
  "benchmark": {
```

```
"benchmark": {
  "ticker": "SP500",
  "name": "S&P 500",
  "currency": "USD",
  "market": "US"
}
}
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> code src/main.py
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> code src/main.py
PS C:\Users\guyma\labs\esilv-marketpulse-g07-Tryhaders> cat src/main.py
from pathlib import Path
import csv
import json

DATA_DIR = Path("data/sample")
LOOKBACK_LABEL = "1 month"
INTERVAL_LABEL = "Daily"

def load_instruments():
    with open(DATA_DIR / "instruments.json", encoding="utf-8") as file:
        return json.load(file)

def load_prices():
```